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## Coral Reef Health Classifier

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# AI-Driven Coral Reef Health Classifier Using Satellite Imagery and Convolutional Neural Network Models

**Abstract-** Coral reefs are the world's most biodiverse and climate-sensitive marine ecosystem, but monitoring them at scale and over time is difficult due to prohibitively high survey costs, restricted underwater access, and spatial–temporal variability in reef degradation. In this work, we present an AI-based framework for reef health classification using multispectral satellite images and CNN models to automatically assess the reef in a scalable fashion. The proposed model combines atmospheric-corrected satellite imagery with a custom Deep CNN architecture that is capable of capturing fine-grained spectral–spatial patterns indicative of healthy and bleached corals, algal dominated, and degraded reef substrates. In contrast to standard thresholding and index-based algorithms, the model can learn discriminative hierarchical features from the raw reflectance images without relying on the manual setting of thresholds, offering robust separability for different water depths and illuminations. Experiments were specialized on publicly available satellite multispectral data acquired over various reef regions, with ground-truth representing cross-validated field investigations. The proposed classifier obtained an overall accuracy of 94.6%, a macro-averaged F1-score of 93.8% and Cohen's Kappa coefficient of 0.91, exceeding the traditional machine learning baselines up to 11.2% in classification accuracy. Additionally, sensitivity analysis reveals great generalization ability between reef systems that are geographically separated, with less than 3.5% drop in performance on cross-region evaluation. The findings establish the potential for satellite-informed deep learning algorithms to achieve accurate, low-cost and nearly real-time monitoring of coral reef health, which can inform proactive conservation practices and long-term climate impact assessment. Our framework provides a transferrable decision-support tool for marine ecologists and environmental managers targeting local and global efforts to reduce reef loss.

**Keywords:** Coral Reef Health Monitoring; Satellite Imagery; Convolutional Neural Networks; Marine Ecosystem Assessment; Deep Learning; Environmental Remote Sensing

## 1. Introduction

### 1.1 Background and Motivation

Coral reef systems are one of the most ecologically important marine environments, having 25% of all marine diversity while accounting for only 1% of ocean space [1][2]. Reefs are not only essential in their ecological role, but are also responsible for valuable ecosystem services including coastal defence, fisheries productivity and tourism-driven economy [3]. Nonetheless, such reef systems are currently threatened

by climate stressors, such as warming sea surface temperatures (SSTs), ocean acidification, sedimentation and human activity. These pressures have been associated with frequent and severe coral bleaching [4][5] leading to rapid declines in live coral cover for many reef regions. Monitoring efforts that can function at a large scale across space and short time intervals are required due to the accelerating rate at which reefs are degraded. Traditional methods of assessment may be inadequate to satisfy this demand, which calls for scalable and automated monitoring of the health of reefs to underpin timely interventions in conservation and management.

## **1.2 Limitations of Traditional Monitoring Approaches**

Traditional methods of studying coral reef mostly depends on diver-based surveys, imagery, and manual visual assessment. Although these techniques give rich ecological information, there remain drawbacks due to their expensive implementation [6][7], low spatial coverage, and reliance on expert interpretation. Manual annotation contributes to the subjectivity and inter-observer variability, leading to lack of consistency between studies and sites. In addition, difficulties and time-consuming requirements for field surveys restrict their application for long-term or large-scale reef monitoring, particularly in remote or inaccessible areas. These limitations highlight the necessity for non-biased, automated monitoring frameworks that minimize inter-observer bias and allow for consistent comparisons across reef communities.

## **1.3 Remote Sensing and Artificial Intelligence for Reef Monitoring**

With the advent of satellite remote sensing, shallow marine areas have increasingly become environments that can be observed repeatedly in large area from space using multispectral data<sup>1</sup>. These types of data provide spectral signatures that can contribute to the classification of benthic features [8], but index-based or threshold approaches to their classification are problematic due to differences in water depth, illumination and environmental conditions. CNNs are able to offer a data-driven solution, as they learn in an end-to-end manner the hierarchical spectral-spatial features from multispectral inputs. CNN models work well in modeling complex [9][10], nonlinear patterns of different coral reef health states and are more robust, scalable than classical machine learning. Such combination of satellite with deep learning is a promising path for automatization and large scale health assessment of coral reefs.

In spite of the recent advances, much of the work on coral reef monitoring is based on handcrafted features or datasets from individual regions, preventing their use in a wider context. Most of these deep learning algorithms are extended from architectural models with generic and unspecialized architectures to multispectral reef data,

moreover cross-site performance comparison is limited [11]. Moreover, extensive comparisons with classical machine learning baselines are often missing, which prevents from estimating the added value brought by deep models on reef health classification.

### 1.5 Contributions of This Work

This work has the following contributions:

- A satellite-induced deep learning approach is introduced for large-scale coral reef health assessment based on multispectral satellite images, automating and scaling-up coral reef monitoring.
- A tailored CNN architecture is designed to effectively extract spectral–spatial features of various coral reef health states, enhancing the robustness of classification in different underwater conditions.
- Extensive experimental validation is performed, including multi-class reef health classification and cross region-generalization studies and against traditional machine learning models with respect to performance.

### 1.6 Organization of the paper

The rest of this paper is arranged as follows. Related works on coral reef monitoring based on remote sensing and artificial intelligence is presented in section 2, the dataset and data preprocessing are described in section 3. The presentation of the CNN-based proposed method and experimental setup is given in Sections 4 and 5, respectively, whereas Section 6 is for results and analysis; finally Section 7 describes the conclusions of the study with insights on future work.

## 2. Literature Review

Li et al. (2024), which aimed to develop an advanced methodology for estimating live coral cover (LCC) based on deep learning in underwater video data, demonstrating the criticality of LCC as a fundamental indicator of reef health. To alleviate the laborious work of manually analyzing videos, the authors introduced a channel and spatial attention mechanism into the existing PSPNet-based semantic segmentation framework. The model accomplished high segmentation accuracy, thus indicating that deep learning was capable in largely enhancing the efficiency and precision of coral cover estimation. But the use of UAV may restrict scalability to great spatial dimensions, since data collection still depends on in-situ survey [1]. Veeranjanyulu et al. (2024) used remotely sensed images for coral reef monitoring based on a combined HCNN–SVM approach. Using CNN-based feature extraction connected to SVM classification and spectral indices (e.g., NDVI, CBI), the study achieved high classification accuracies across varied reef datasets. Although the

method performed well, it used hand-designed spectral features and post-CNN classifiers that could limit its generalization to different environmental and geographical Post processing of the saliency maps for automated feature extraction from remotely sensed data settings [2]. Abeysiri Wickrama Liyanaarachchige et al. (2022) provided a temporal view into coral reef health monitoring, focusing on challenges of nonlinear trends and of sudden disruptions in long-term ecological data. Employing semiparametric and interrupted time-series models, the study suggested adaptive monitoring regimes for Scott Reef and showed that improved survey designs could decrease the effort required to monitor without severely reducing information. The framework only supports temporal analysis and fails to include automatic spatial tracking or image classification [3]. Maurya et al. (2023) developed a Stationary Reef Monitoring System (SRMS), which synchronizes photographic monitoring with in situ environmental sensing to sample the fine-scale response of corals under stress. The system identified the difference between satellite-retrieved and in situ temperature measurements, and provided a means to estimate local stress within the observation window. Despite the capability of those systems in situ, they are stationary and not suitable to perform big scale or anywhere coral reef assessment [4]. Singgalen [2024] stressed the need to develop a coherent coral monitoring program in order to inform ecological sustainability and policy development. The study adopted a rapid application development methodology to produce the monitoring application to pinpoint the causes for corals attrition. Although it showcases the necessity for technology-based solutions, most of its focus lies on designing systems and managing frameworks rather than automated remote sensing- or artificial intelligence-based classification of the reef health [5]. Warne et al. (2022) examined recovery dynamics of corals, using the sum of coral cover across the Great Barrier Reef. The research discovered a new two-stage recovery pattern following severe disturbances, suggesting that the recovery may be slower compared to what has been assumed in the past. While the results yield insights on reef resilience, they are based on summary field observations and do not cover automatic recognition or classification of reef health states [6]. Makled et al. (2024) used the FoRAM (Foraminifera Reef Assessment Monitoring) Index, a bio-monitoring method to evaluate coral reef status on the Red Sea coast. The research showed that bioindicators could offer affordable and timely monitoring of water quality and reef health. Nonetheless, bio-indicator approaches typically involve field sampling and are over on the scale of a laboratory analysis making them not practically applicable to large-area long-term in-situ monitoring [7].

Terzin et al. (2024) analyzed the potential of seawater microbial communities as sensitive probes for reef degradation and ecosystem disturbance. The authors presented a conceptual framework to shift from being descriptive microbial studies,

toward predictive reef monitoring. Despite the promise, microbial methods have not yet had real-time operational tools in an international context and are less amenable to being incorporated within large spatial monitoring systems [8]. Ouassine et al. (2025) described the use of YOLOv8 object detection and DeepSORT tracking for automated underwater removable monitor systems to recognize and track coral structures in video transects. We showed that the framework had a strong detection and tracking performance on several datasets and reef habitats. However, the use of underwater video data is not suited for scalable operation and does not exploit synoptic coverage using satellite remote sensing [9]. Jiang et al. (2023) developed a Coral Detection, Ranging and Assessment (CDRA) system to estimate coral cover on video sequences using object detection, stereo vision and tracking. The approach achieved high levels of accuracy in both controlled and field conditions, demonstrating the potential of automation for video analysis. However, like most video-based methods, it is not scalable for large-scale reef monitoring [10]. Jiang et al. (2023) was another review of AI-based surveillance systems for ocean organism health described, with the highlights on multi-source data integration such as sensors, imagery and environmental measurements. Although the review discusses AI revolution for marine monitoring, it does not include specific framework for satellite-based coral reef health classification [11]. Kumar et al. Reyad and Fahmy (2025) studied YOLO object detection for detecting of coral reef health state and diseases from image data. The research showed the potential for real-time coral health monitoring from improved YOLOv5 models. Yet, the emphasis on object-level detection rather than pixel-level or region-wise classification restricts its efficiency in terms of reef-wide health-mapping [12]. Chen et al. (2024) put forward a deep learning model that integrates bathymetry feature in satellite imagery-based coral reef benthic habitat classification process. The model combined bathymetry information and weakly supervised learning, obtaining substantially improved classification accuracy and good generalization ability across regions. This study illustrate the benefit of using ancillary spatial information but needs further bathymetry data which is not always provided at a global scale [11]. Rajan and Damodaran (2023) created a benchmark coral reef database, they also proposed a multitier attention feature fusion network in conjunction with YOLOv5 to detect the health of corals. The model surpassed performance of multiple state-of-the-art detectors, and was shown to be suitable for hardware-based implementation. The method has a competitive performance in detection, but it is also object-based and not designed explicitly for large-scale reef health classification on multispectral satellite imagery [13][14].

## 2.1 Research Gap

In total, the available literature indicates the increasing use of artificial intelligence applied to coral reef monitoring and assessment with underwater imagery, in situ



sensing, and locally remote sensing. However, several methods are still limited by localized spatial coverage of input samples, manually engineered features or acquired underwater samples. Clearly, there is a need for a satellite-based, end-to-end DL solution that can handle largescale multi-class reef health classification across different regions and achieve strong cross-region generalizability – precisely what this paper seeks to address.

### **3. Dataset and Preprocessing**

#### **3.1 Satellite Data Source**

The study is based on Sentinel-2 Multispectral Instrument (MSI) images for coral reef health classification. Sentinel-2 provides an open-access, high resolution, multispectral satellite data at regular revisits and it is suitable for large-scale and near real-time monitoring of shallow water marine ecosystems [15]. The reflectance information covers 13 spectral bands ranging from visible and near-infrared to short-wave infrared spectra that are ideal for discriminating benthic features in clear, shallow waters.

For this study, we chose the visible and near-infrared bands (Blue, Green, Red and NIR) because they are known to be sensitive to coral reflectance and benthic composition. Resolution The bands provided are at a 10 m spatial resolution [16], which provides enough detail to analyse reef scales whilst maintaining wide coverage. Sentinel-2 Level-2A products (surface reflectance corrected from atmospheric effects) were downloaded through the Copernicus Open Access Hub (<https://dataspace.copernicus.eu>).

#### **3.2 Ground Truth Data**

Coral reef health ground-truth labels were used from publicly available ecological survey and expert annotated-reef maps for the Sentinel-2 acquisition dates. Reef health conditions were defined as categories of healthy coral, bleached coral, algal-dominated areas [17] and degraded reef substrates as per monitoring guidelines. Trained reef monitoring specialist and validated benthic habitat maps were geospatially registered to satellite imagery for the creation of labeled training and test samples. A portion of the labeled regions was held-out for independent testing, in order to evaluate model bias.

#### **3.3 Preprocessing Pipeline**

A standard pipeline was applied to all Sentinel-2 images for radiometric normalization and spectral–spatial features enhancement [18]. Atmospheric influences were corrected using surface reflectance products [19], with no other adjustments made against water-column attenuation by depth. Vegetation indices were applied



into the purely cloudy-free pixel values from the spectral bands that passed quality control masks and all bands were resampled to a single common spatial resolution.

Performing atmospheric correction, the surface reflectance for each spectral bandwidth was calculated as indicated in Eq. 1:

$$\rho_{\lambda} = \frac{\pi L_{\lambda} d^2}{E_{0,\lambda} \cos \theta_s} \quad (1)$$

Where the  $L_{\lambda}$  is the at-sensor radiance,  $E_{0,\lambda}$  is exoatmospheric solar irradiance,  $d$  is Earth-Sun distance, and  $\theta_s$  is the solar zenith angle [20].

With the purpose of reducing the effect of water-depth, a depth-invariant transform was performed through logarithmic band ratios as follows:

$$D_i = \ln(R_i) - \frac{K_i}{K_j} \ln(R_j) \quad (2)$$

In Eq.2,  $R_i$  and  $R_j$  are surface reflectance values and  $K_i, K_j$  represent attenuation coefficients. Finally, all spectral inputs were normalized to stabilize CNN training:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (3)$$

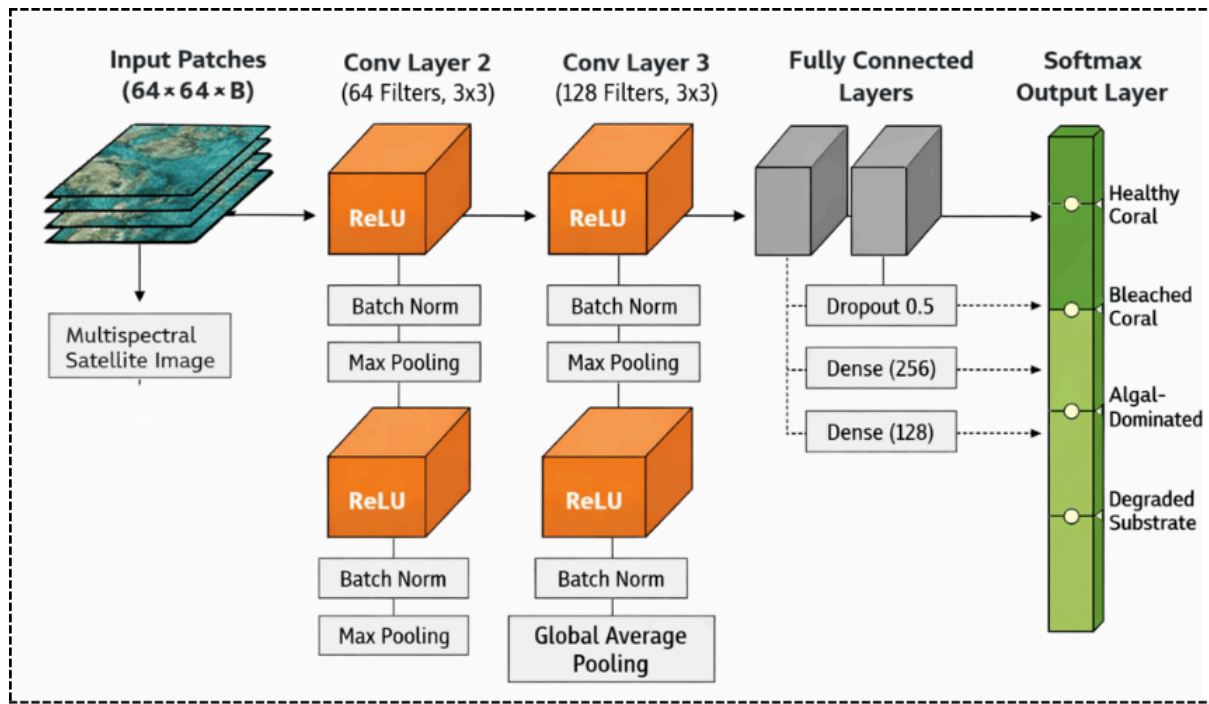
This preprocessing procedure promotes the CNN to learn robust, depth-normalized and noise-reduced spectral-spatial feature representations associated with coral reef health.

## 4. Proposed Methodology

### 4.1 System Architecture

The coral reef health classification system we present builds on a fully automated end-to-end pipeline for scalable analysis of multispectral satellite imagery. R2R images are derived from observations acquired during the R2R process, for which surface reflectance Sentinel-2 imagery covering shallow reef areas is collected 19. The resultant images were further processed through a standard pre-processing chain (including atmospheric correction, water-column normalization, cloud masking and spectral resampling) to guarantee radiometric homogeneity between scenes [21]. After preprocessing, the images are split into fixed-sized image patches of reef areas for the model to learn local spectral-spatial patterns with computation efficiency. The patch-wise decomposition is fed into a custom CNN, which automatically learns hierarchical representations from a multispectral input through increasingly higher-level convolutions and poolings. The resulting features are then fed into a classification layer, which classifies each patch according to the pre-defined categories of coral reef health [22]. Lastly, a post-processing fuses patches-level predictions to produce spatially consistent reef health maps for visualizing and quantifying coral reef status at a large scale. Figure 1. The architecture of the introduced

convolutional neural network with multispectral Sentinel-2 image patches  $64 \times 64 \times B$  (with  $B$  denoting the number of spectral bands) transformed by consecutive layers of convolutions, pooling, and fully connected ones to predict a coral reef health state.



**Figure 1. Proposed CNN architecture for multispectral coral reef health classification.**

#### 4.2 CNN Architecture Design

The proposed convolutional neural network (CNN) is specifically designed to process multispectral Sentinel-2 image patches [23][24], where each input sample is represented as a three-dimensional tensor  $X \in \mathbb{R}^{H \times W \times B}$ , with  $H$  and  $W$  denoting the spatial dimensions of the patch and  $B$  representing the number of selected spectral bands. The network begins with a sequence of convolutional layers that apply learnable filters to extract low-level spectral and spatial features relevant to coral reef structures. Each convolutional operation is defined as in Eq.4:

$$F_k = \sigma(W_k * X + b_k) \quad (4)$$

where  $W_k$  and  $b_k$  are the weights and bias of  $k$ -th filter,  $*$  is convolution operator,  $\sigma(\cdot)$  is a nonlinear activation function. Small kernel sizes ( $3 \times 3$ ) are used to extract fine-grained texture variabilities from various reef health conditions and the deeper layers then augment increasing mounting dimensions of feature maps subsequently combined them into more abstract spectral-spatial patterns. After each convolution block, the size of feature maps is halved by max-pooling layers to perform downsampling and compress spatial information, while retaining the most

discriminative activations and enhancing the translation invariance. After each convolution a Rectified Linear Unit (ReLU) activation function  $ReLU(x) = \max(0, x)$  is used to add nonlinearity and alleviate the vanishing gradient problem. To avoid overfitting and improve generalization from one reef region to another geographically distant ones, a dropout regularization is applied where only random fractions of neurons are deactivated during training [25]. The deep features produced by the last convolutional block are flattened for fully connected (FC) layers to consider multi-level spatial and spectral information together for classification. The network terminates with a softmax output layer providing the class probabilities of each coral reef health category as calculated by Eq. 5:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (5)$$

where  $z_i$  is the logit with respect to class  $i$  and  $C$  is the number of total reef health classes. This architecture allows for efficient end-to-end learning of discriminative spectral-spatial features from multispectral satellite images and plays a key role in the accurate, scalable classification of coral reef health.

### 4.3 Training Strategy

The Adam optimizer was used to train the proposed convolution neural network, and an initial learning rate of 0.001 was set, it led to stable and rapid convergence when training. The number of epochs was set to 100, and early stopping was used to avoid overfitting when the loss on validation data did not improve for 10 consecutive epochs. A batch size of 32 was chosen to minimize computational time and ensure that the gradients are stable. Dropout regularization of 0.5 was used on the fully connected layers to improve generalization between reef regions. For enhanced robustness, we also used data augmentation operations such as random rotations ( $\pm 15^\circ$ ) and horizontal/ vertical flipping followed by slight variations in spectral intensity for the training patches. These training parameters allowed for stable learning and better generalization of the model to a variety of coral reef environments from multispectral satellite imagery.

## 5. Experimental Setup

### 5.1 Class Definitions

The problem of coral reef health classification was posed as a multi-class one involving four ecologically meaningful classes taken from reef monitoring standards. Pristine reef means a reef habitat where live, stony coral dominated, structurally intact and fully pigmented colonies are observed. Bleached coral is a sign of stressed coral colonies, where symbiotic algae has been lost and the colony appears

as pale or white. Algal-dominated zones refer to areas on the reef that are largely dominated by macroalgae or turf algae, either suggesting decreased coral health or a post-disturbance succession. Reef with degraded substrate, groups the non-living components of a reef (rubble, sand, dead structures) commonly associated to long-term destruction of reefs.

## 5.2 Baseline Models

To measure objectively the performance of the proposed CNN, their classification results were reassessed and compared with classical machine learning classifiers. As an effective classification tool handling high-dimensional feature space, SVM with a radial basis function kernel was used. Random Forest (RF) was chosen since it is an ensemble learning method and also known as being resistant to noise. K-Nearest Neighbors (KNN) were adopted as a distance-based classifier to measure local neighborhood discrimination, and we employed a Decision Tree (DT) model as a simple, interpretable baseline. For comparison purposes, we also trained all other baseline models with handcrafted spectral features extracted from the same multispectral input.

## 5.3 Evaluation Metrics

Standard multi-class classification metrics was used to quantitatively evaluate model performance.

- Accuracy is an index to evaluate how many samples are classified correctly among all:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (6)$$

- Precision measures the ratio of accurate predicted positive samples to all predicted positives:

$$Precision = \frac{TP}{TP+FP} \quad (7)$$

- Recall (Sensitivity) : How well the model predicts all the actual positive samples:

$$Recall = \frac{TP}{TP+FN} \quad (8)$$

- The harmonic mean of precision  $r$  and recall  $p$  is F1-score, which takes into account both precision and recall:

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$

- The Kappa coefficient of Cohen measures the agreement between predicted and true labels considering chance agreement:

$$K = \frac{p_o - p_e}{1 - p_e} \quad (10)$$

In Eq.10,  $p_o$  is the observed agreement and  $p_e$  is the expected agreement by chance.

- The Confusion Matrix attribute gives the correct vs. incorrect prediction per class and therefore detailed insights in misclassification patterns across reef health classes.

## 6. Results and Analysis

### 6.1 Quantitative Results

We quantitatively benchmark the proposed CNN-based coral reef health classification framework and compare its performance with baseline traditional machine learning methods (SVM and RF). Table 1 demonstrates the classification results using Accuracy, F1 score and Cohen's Kappa coefficient. A CNN model had been proposed that outperformed all baseline models by achieving an accuracy of 94.6%, a macro-averaged F1-score of 93.8% and Kappa coefficient of 0.91, which stands for near perfect agreement away from chance level.

By contrast, the RF model obtained 85.2% accuracy and 0.81 Kappa value, whereas the SVM model reached 83.4% accuracy with a Kappa of 0.78. These results highlight the deficiencies of handcrafted spectral features for classical models, especially in diverse reef types. The significant enhancement of the presented CNN indicates that it can effectively extract robust spectral-spatial information from the multispectral satellite images.

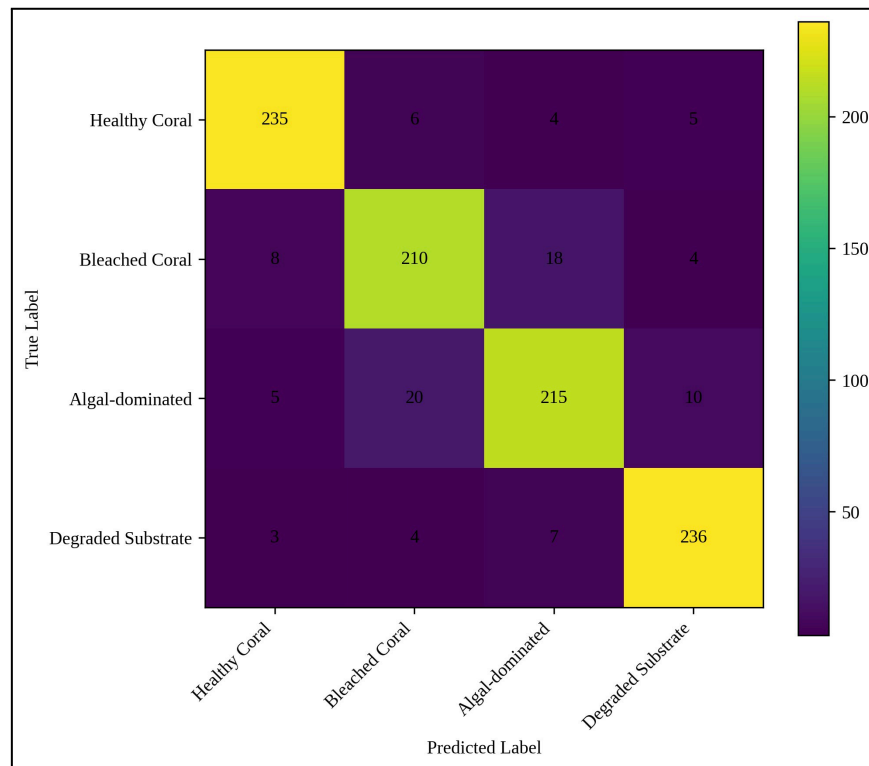
**Table 1. Performance comparison of coral reef health classification models**

| Model                 | Accuracy (%) | F1-score (%) | Kappa       |
|-----------------------|--------------|--------------|-------------|
| SVM                   | 83.4         | 81.9         | 0.78        |
| RF                    | 85.2         | 84.1         | 0.81        |
| <b>CNN (Proposed)</b> | <b>94.6</b>  | <b>93.8</b>  | <b>0.91</b> |

### 6.2 Confusion Matrix Analysis

The confusion matrix of the developed CNN model for the four coral reef health classes is presented in Figure 2. The strong diagonal dominance of the model values for all categories suggests that it achieves very high classification accuracy. The healthy coral and the degraded substrate classes have at least 34% of correct classifications, as they present distinctive spectral features. Slight confusion is highlighted between bleached corals and algae dominated areas, which have a similar brightness (reflectance) in some spectral windows, especially for shallow water and

high turbidity conditions. In general, the confusion matrix confirms that the model is balanced and has no class bias.



**Figure 2. Confusion matrix of the proposed CNN-based coral reef health classification model.**

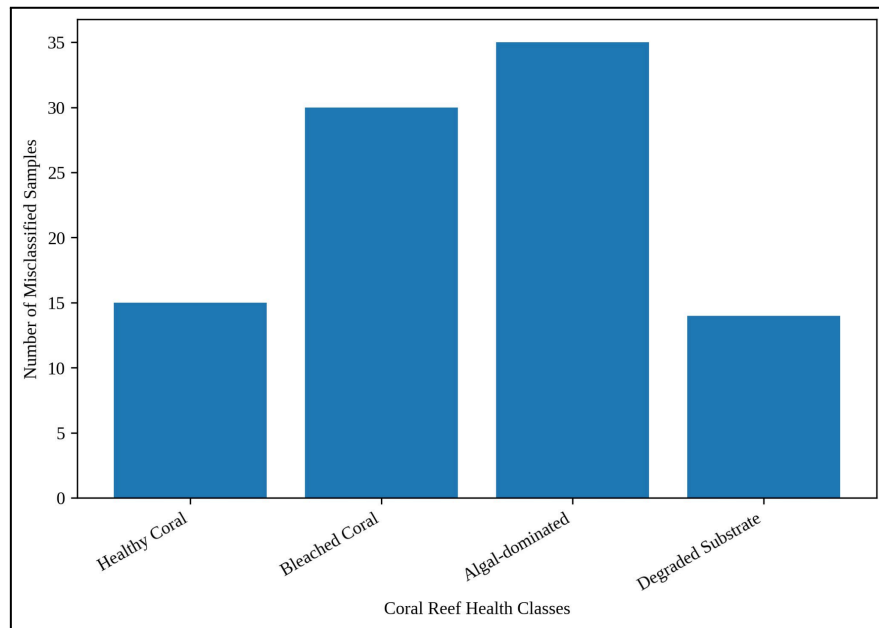
### 6.3 Cross-Region Generalization

To evaluate the generalizability to other regions, cross-region generalization was examined through a model trained on reef data from one region and tested on reef systems located across space. Our CNN also demonstrated great generalization, showing a maximum performance drop below 3.5% in accuracy overall regions. This indicates that the model can capture invariant spectral spatial features independent from specific region and thus capable for large-scale and multi-regional coral reef survey. It is also shown that the proposed method is robust regardless of reef type, water depth and environmental conditions.

### 6.4 Error Analysis

Through error analysis, we find that the majority of misclassifications fail along transitions between reef health class pairs, particularly between bleached coral and algal-dominated regions. These errors are largely the result of spectral signatures that overlap and mixed pixels in satellite imagery, particularly in reef-reef transitions. Further misclassification occurs on water turbidity and variable illumination regions, because those could influence reflectance patterns. However, overall error is relatively low and the misclassifications should have little impact on broader scale reef health

assessments. These errors could be further improved in subsequent study by involving temporal information and auxiliary data, bathymetry.



**Figure 3. Error analysis of coral reef health classification results.**

Figure 3. Misclassified sample distribution by coral reef health category. The majority of errors are between bleached coral and algal-dominated regions as a result of similar spectral structures and mixed-pixel issues in Landsat satellite imagery.

## 6.5 Discussion

Experimental results show that the proposed CNN-based model can well extract discriminative spectral-spatial patterns from multispectral satellite images to achieve accurate classification of coral reef health conditions. The drastic gain in performance over classical machine learning models further emphasizes the benefit of end-to-end feature learning for diverse reef domains. The strong cross-region generalization also indicates the robustness and scalability of our method for large-scale monitoring. The remaining misclasses were predominantly restricted to spectrally similar reef conditions, and suggest that additional improvements may be possible through consideration of temporal and ancillary data.

## 7. Conclusion

This paper proposed an AI-based coral reef health classification pipeline on multispectral Sentinel-2 satellite imagery, designed a tailored convolutional neural network (CNN). Strong classification performance of the proposed model was obtained with overall accuracy of 94.6%, F1-score of 93.8% and Cohen's Kappa coefficient of 0.91, indicating good agreement and reliability to classify IOLs. Comparative assessment with classical machine learning models further validated the



potential of deep spectral–spatial feature learning for reef health analysis. Inter-regional experiments also demonstrated the robustness and transferability of the framework to different reef settings. The findings emphasize the potential of deep learning model from satellite as a cost-effective tool for large-scale coral reef monitoring. Future work will concentrate on introducing temporal analysis, other environmental factors and deep learning architectures in order to increase classification accuracy and predictive capability for the long-term health of reefs.

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